

Breathing and Exchange of Gases

Topic - Full Chapter

1. Given below are two statements :

Statement-I : Tracheal rings are supported by bones.

Statement-II: Trachea is a straight tube extending up to mid thoracic cavity.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

2. The mucus produced by the goblet cells

- (a) moistens the incoming air
- (b) trap the fine dust particles
- (c) filter out large dust particles
- (d) warms the incoming air

- (1) a , b , c are correct.
- (2) only a is correct.
- (3) b and d are correct.
- (4) a and c are incorrect.

3. Contraction in diaphragm causes increase in thoracic volume in the ____.

- (1) dorso-ventral axis
- (2) antero-posterior axis
- (3) dorso-posterior axis
- (4) antero-ventral axis

4. Select correct statement for transport of oxygen?

- (a) seventy percent of oxygen is carried as bicarbonates.
- (b) about ninty seven percent of oxygen is transported by RBCs in blood.
- (c) it is carried by haemoglobin as carb-amino haemoglobin.

- (1) a, b
- (2) b, c
- (3) only b
- (4) only c

5. Match List-I with List-II

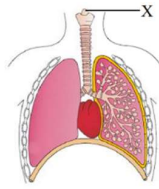
List – I	List - II
I. Earthworm	A. Moist cuticle
II. Aquatic arthropods	B. Gills
III. Cockroach	C. Lungs
IV. Birds and reptiles	D. Tracheal tubes

- (1) I-B, II-A, III-D, IV-C
- (2) I-A, II-D, III-C, IV-B
- (3) I-A, II-B, III-D, IV-C
- (4) I-A, II-B, III-C, IV-D

6. Lungs are situated in;

- (1) Abdominal cavity
- (2) Thoracic cavity
- (3) Haemocoelomic cavity
- (4) Pleural cavity

7. what is the function of structure marked as X ?



- (1) To prevent the entry of food into larynx.
- (2) To filter and warm the air.
- (3) To help in exchange of gases.
- (4) To catch dust and bacteria.

8. Which is the correct sequence of air passage in man?

- (1) External Nostril → Nasal passage → Internal nostril → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli
- (2) Nose → Larynx → Pharynx → Bronchioles → Bronchi → Alveoli
- (3) Nose → Pharynx → Trachea → Larynx → Bronchi → Bronchioles → Alveoli
- (4) Nose → Larynx → Bronchi → Pharynx → Trachea → Bronchioles → Alveoli

9. Given below are two statements :

Statement-I : A healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute.

Statement-II: Various pulmonary capacities, can be used in clinical diagnosis.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

10. Given below are two statements :

Assertion (A): Asthma is difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

Reason (R): In asthma there is damage of alveolar walls and reduction of respiratory surface.

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, but Reason (R) is false.
- (4) Assertion (A) is false, but Reason (R) is true.

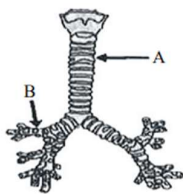
11. At the alveolar site where different CO₂ compounds like carbamino compound, bicarbonates etc. are dissociated and release the CO₂ . Find out the suitable factor which is responsible for dissociation for CO₂-compounds?

- (1) PCO₂ is high
- (2) PCO₂ is low
- (3) Bohr's effect
- (4) High H⁺ concentration

12. Select the correct option for expiration?

- (1) expiration is a passive process.
- (2) during this process diaphragm and sternum returns to their normal position.
- (3) intrapulmonary pressure is slightly higher to the atmospheric pressure.
- (4) all of the above.

13. In the given figure, label A represents _____ while label B represents _____.



- (1) A-Trachea, B-Bronchus
- (2) A-Alveolus, B-Bronchiole
- (3) A-Bronchiole, B-Trachea
- (4) A-Trachea, B-Bronchiole

14. Consider the following statements

Statement-I : Haemoglobin is a red coloured iron containing pigment present in the RBCs.

Statement-II: O_2 can bind with haemoglobin in a irreversible manner to form oxyhaemoglobin.

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

15. Which of the following statement is not true?

- (1) The partial pressure of oxygen in deoxygenated blood is 40 mm Hg.
- (2) The partial pressure of oxygen in oxygenated blood is 95 mm Hg.
- (3) The partial pressure of oxygen in the alveolar air is 104 mm Hg.
- (4) The partial pressure of carbon dioxide in deoxygenated blood is 40 mm Hg.

16. Read the following statements:-

- (A) It is a chronic disorder
- (B) Major cause of it is cigarette smoking.
- (C) Alveolar walls are damaged
- (D) Respiratory surface is decreased.
- (E) It is due to inflammation of bronchi & bronchioles.

How many of the above statements are correct about "emphysema" disorder ?

- (1) Two
- (2) Three
- (3) Four
- (4) All

17. Match List-I with List-II to find out the correct option.

List - I

List - II

- | | |
|----------------|------------------------------|
| I. TV | A. 1000 mL to 1100 mL |
| II. IRV | B. 500 mL |
| III. RV | C. 1100 mL to 1200 mL |
| IV. ERV | D. 2500 mL to 3000 mL |

- (1) I-A, II-B, III-D, IV-C
- (2) I-C, II-A, III-D, IV-B
- (3) I-B, II-D, III-C, IV-A
- (4) I-C, II-D, III-A, IV-B

18. Among vertebrates, fishes use _____ whereas reptiles and birds respire through _____.

- (1) skin, lungs
- (2) gills, cuticle
- (3) gills, lungs
- (4) gills, spiracles



19. Given below are two statements :

Assertion (A): The lungs are situated in the thoracic chamber which is anatomically an air-tight chamber.

Reason (R): The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung (pulmonary) cavity.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, but Reason (R) is false.

(4) Assertion (A) is false, but Reason (R) is true.

20. Match List-I with List-II to find out the correct option.

List - I

List - II

I. Expiration

A. Forceful expiration

II. EICM

B. Inhalation

III. IICM

C. Exhalation

IV. Inspiration

D. Contract during

normal inspiration

(1) I-A, II-B, III-D, IV-C

(2) I-C, II-A, III-D, IV-B

(3) I-A, II-D, III-C, IV-B

(4) I-C, II-D, III-A, IV-B

ANSWER KEY

1. (3) 2. (1) 3. (2) 4. (3) 5. (3) 6. (2) 7. (1) 8. (1) 9. (1) 10. (3)
11. (2) 12. (4) 13. (1) 14. (2) 15. (4) 16. (3) 17. (3) 18. (3) 19. (1) 20. (4)

